

STEPGEN · PROBLEM STATEMENT

StepGen problem statement

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.4	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (SGN-DDR-001) on budget, legal route and co-design partners; SwapCell interface v0.3; numbers checked against SGN-CAL-001
0.3	2026-09-25	Amish Chadha	Concept changed at Amish's direction from a stationary stepper generator to a walking-treadmill vehicle
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work, honest energy framing)
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepGen problem statement

Many people who would benefit from an electric bike never use one because riding needs cycling skill, confidence on pedals and a seated posture that does not suit everyone. Walking is the one way of moving that almost everyone already knows. StepGen asks whether a small electric vehicle can be driven by walking: the rider stands upright on a short treadmill belt between the wheels, walks at an ordinary pace, and the vehicle moves forward at e-bike speed.

Concept change. Amish decided on 2026-09-24 that StepGen is a walking-treadmill vehicle in the manner of the Dutch Lopifit walking bike, not a stationary stepper generator. The earlier generator concept, which charged a PowerBox, is dropped entirely. Version 0.2 of this document describes the earlier concept and remains in Git history. On 2026-09-25 Amish accepted the TRL 2 recommendations, including a \$650 budget, the route for the legal question and the move to SwapCell interface v0.3 (SGN-DDR-001). The same day he accepted the TRL 3 recommendations, including a salvage reference build that keeps the \$650 budget (SGN-DDR-002).

■ The problem

Short trips of 2 to 15 km (about 1 to 9 mi) to work, school, a market or a clinic are too long to walk comfortably and too short or too costly for a car or a taxi. E-bikes fill this gap well for people who already cycle. They leave out:

- people who never learned to ride, which is common among adults in many countries and among women in some communities;
- older adults and people with some balance or joint limits who are nervous on a saddle and pedals;
- people whose clothing, such as long skirts or wraps, makes a diamond-frame bike awkward;
- people for whom a seated, pedalling posture is uncomfortable, for example after hip or knee problems, while ordinary walking is fine.

Walking needs no new skill. On a walking vehicle the rider stands, holds a handlebar and walks. The walking motion is a natural control input: walk and the vehicle goes, stop walking and it slows. The rider gets light exercise at a walking effort while covering distance at bicycle speeds.

■ Why the belt is a control input, not the engine

A person walking puts only a small horizontal force into the ground, because walking is mostly about supporting and moving body weight, not pushing backward hard. On a free-running belt over a roller bed this is about 22 to 30 N (5 to 7 lbf) at 5 km/h, or roughly 30 to 41 W of mechanical power (estimate, see SGN-CAL-001). This vehicle at 20 km/h on the flat needs about 136 W at the wheel with an 80 kg rider. A purely mechanical belt-to-wheel drive would therefore move at about walking pace and would feel harder than walking, and people who built such drives report that they are slower than simply walking ([SolidSmack](#), "Walk Don't Ride").

StepGen is framed around that fact. The belt tells the vehicle that the rider is walking and how fast; an electric hub motor supplies most of the propulsion; and a shared SwapCell battery pack supplies the energy. This is also how the Lopifit works: its 250 W motor drives the vehicle, and the belt does not recharge the battery ([New Atlas](#)).

■ Users and context

User	Need	Context
Adults who do not cycle	Cover 2 to 15 km trips without learning to ride	Towns and peri-urban areas with cycle lanes or quiet roads
Older adults and people with mild balance or joint limits	An upright, familiar movement with a handlebar to hold	Short errands, clinic visits; must be able to step off easily
Commuters who want light exercise	Arrive without sweating, get some walking in	Flat to rolling urban routes, 5 to 15 km each way
Riders who share batteries across portfolio vehicles	One pack that also runs SunSpoke, WaterWalker assist and a PowerBox	Households or groups with a SwapCell dock
Local builders and bike mechanics	A frame and parts they can build, fix and adapt	Garage or market-town workshop with a welder

■ Constraints

- Garage-buildable prototype with a budget of \$650 USD in `project.yaml` (decided by Amish, 2026-09-25), excluding the SwapCell pack, which is priced once in the SwapCell project. Amish decided on 2026-09-25 (SGN-DDR-002) that the reference build uses a salvaged walking-pad treadmill and a donor 20 in bike, which prices at about \$546; an all-new build (about \$721) is the fallback (see SGN-CAL-001).
- Uses the shared SwapCell pack (interface v0.3: 13S lithium-ion, about 46.8 V nominal, 39.0 to 54.6 V, about 468 Wh, 2.85 kg), with a 10 k Ω coding resistor in the INTERLOCK loop for wake and a vehicle receiver to latch class V1. StepGen must not change the interface locally; conflicts go back to the SwapCell project.
- Legal in intent: stay inside pedelec-like limits (250 W rated motor, assist cut before 25 km/h, motor only while the rider walks) so the vehicle can be argued to be as safe and as limited as an e-bike. Whether it legally counts as one is an open question (see below).
- Safe for untrained riders: no reachable pinch points, motor cut-off when the rider stops walking, lets go of a lanyard or pulls a brake.
- Rider can put a foot on the ground easily: low deck, open sides.
- Fits a cycle lane, a lift and a hallway: narrow and not much longer than a long bicycle.
- Every circuit stays below 60 V DC.

Safety: StepGen puts a standing person on a moving belt on a moving vehicle at up to 25 km/h, powered by a 468 Wh lithium-ion pack at up to 54.6 V DC. Falls, pinch points at the belt rollers and rear tire, braking with feet on a belt, pack retention and battery fire are the main hazards; SGN-PRC-001 lists the controls. Nothing in this repo clears the design for building or riding.

■ Legal status

On current rules StepGen does not clearly fit the e-bike category anywhere, because e-bike definitions assume pedals.

- **EU.** Pedal-assisted cycles with a continuous rated motor power up to 250 W, whose assistance is cut before 25 km/h and only while the rider pedals, are excluded from moped type approval ([Wikipedia summary of EN 15194 and Regulation 168/2013](#)). A walking vehicle has no pedals. In the Netherlands, guidance for walking bikes says an electric walking bike may use the pavement only up to 6 km/h and must otherwise use the cycle path or road, and that liability insurance is required for electric versions ([Scouters.nl](#)). Status in other member states is not established here.
- **US.** Federal law defines a low-speed electric bicycle as a two- or three-wheeled vehicle with fully operable pedals and a motor under 750 W with a top motor-only speed under 20 mph ([15 USC 2085](#)). Most states use three classes: class 1 (assist only while pedalling, to 20 mph), class 2 (throttle, to 20 mph) and class 3 (assist only while pedalling, to 28 mph), and all of them require operable pedals ([Redtail, state guide](#)).

The design therefore targets the stricter EU limits. Amish decided on 2026-09-25 that a first target country is picked and the vehicle's legal category confirmed there before any road use. Which country is still open, awaiting Amish.

■ Out of scope

- Generating household electricity. The earlier stepper generator concept is dropped; StepGen does not charge a PowerBox. It draws from a SwapCell pack like the other portfolio vehicles.
- Cargo carrying beyond a small bag or basket.
- Speeds above 25 km/h, or motors above 250 W rated.
- Mains-voltage electronics or a built-in charger. Packs charge in a SwapCell dock.
- Off-road use.

■ Prior work

- **Lopifit (Netherlands).** An electric walking bike with a treadmill belt between a 28 in front wheel and a 20 in rear wheel, a 250 W motor, a 960 Wh battery of about 5 kg, disc brakes front and rear, a top speed of 25 km/h and a claimed range of 50 to 70 km. It is about 2.22 m long, 0.42 m wide and about 55 kg, with a handlebar height of about 1.2 m, and costs from about €2,999 ([Lopifit](#)). Its handles carry sensors that stop the treadmill with a small hand movement. It proves the concept works and is sold, but its price and mass put it out of reach of the users above.
- **Mechanical treadmill bikes.** Belt-driven bicycles without a motor have been built by hobbyists. The reported result is that they are slower than walking because the belt cannot deliver useful torque at low speed ([SolidSmack](#)).
- **Stand-up elliptical bikes (ElliptiGO).** Mechanical, upright, with an elliptical stride that does deliver strong propulsion, but this is a running-like exercise machine for fit users, not a walking vehicle ([Wikipedia](#)). The upright posture also costs more air drag than a crouched cyclist.
- **Kick scooters and e-scooters.** Standing, easy to learn, but e-scooters use a throttle and give no exercise, and small wheels cope poorly with rough surfaces.
- **Portfolio siblings.** SunSpoke (bolt-on e-bike kit) and WaterWalker (walking water carrier with optional assist) share the SwapCell pack and the same 250 W hub motor class.

■ User research and co-design

Design with, not for: this design is for people the author may not be part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO, cycling charity, older-adult group or university). By Amish's 2026-09-25 rule, community designs pick co-design partners per area later, so the partner stays open.
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate trip distance, route, speed, balance, step-off and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design

First-session questions:

- Does walking on a moving vehicle feel safe, and at what speed does it stop feeling safe?
- Can users step on and off, and put a foot down, without help?
- Is two-wheel balance acceptable for non-cyclists, or is a three-wheel version needed?
- Where would the vehicle be kept, and can users lift or wheel it into a building?

■ Open questions

- Which user group to start with: non-cycling adults, older adults, or commuters? Proposed, awaiting Amish (no recommendation). The partner is picked later, per area.
- Can a two-wheeled walking vehicle be balanced by people who never learned to cycle, or does the first prototype need three wheels? To be learned in co-design.
- Which first target country, and what is the legal category of a pedal-less walking vehicle there? The route is decided (confirm before any road use); the country is proposed, awaiting Amish.
- Is a 1.0 m belt long enough for the intended users? SGN-CAL-001 shows it fits a 1.75 m rider only up to 5 km/h. To be learned in co-design.